

Quantum Machine Learning: An Overview

Quantum machine learning (QML) is a burgeoning field at the intersection of quantum computing and artificial intelligence. In recent years, the integration of quantum mechanics with machine learning algorithms has sparked substantial interest among researchers and technologists alike. Here's a quick look at the essentials of creating quantum algorithms for AI models, their practical use cases on open source platforms, and best practices for implementing these advanced algorithms.



Quantum machine learning (QML) combines the principles of quantum computing with machine learning algorithms to create models that can process and analyse large datasets more efficiently than classical computers. The inherent parallelism and entanglement properties of quantum computing enable it to perform complex computations at

unprecedented speeds. This capability is particularly beneficial in the context of machine learning, where large scale data processing and model training are often bottlenecks.

In recent years, advances in quantum computing have made QML a practical reality. Companies like IBM, Google, and Microsoft have developed quantum processors and platforms that are accessible to

researchers and developers. These platforms provide the necessary tools to design, test, and deploy quantum algorithms for various applications, including AI models. The momentum behind QML is driven by its potential to solve problems that are currently intractable for classical computers, such as optimising large datasets, enhancing pattern recognition, and improving predictive models.